

Gebze Technical University

Dep. of Mechanical Engineering

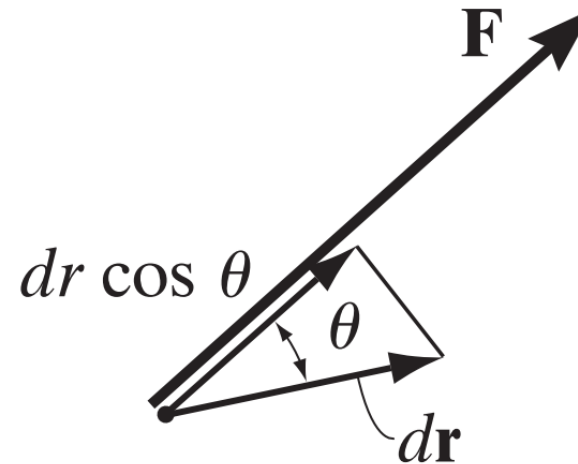
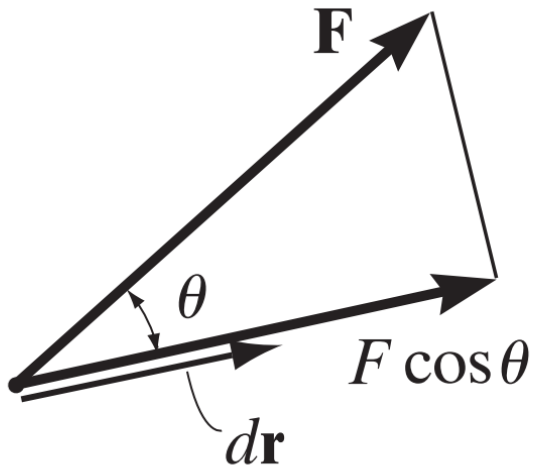
Statics (ME 231)

Lecture 20: Virtual Work and Potential Energy

By Saeed Lotfan

Work of a Force

The work produced by $\mathbf{F}$ is defined as



$$dU = F dr \cos \theta$$

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$$dU = \mathbf{F} \cdot d\mathbf{r}$$

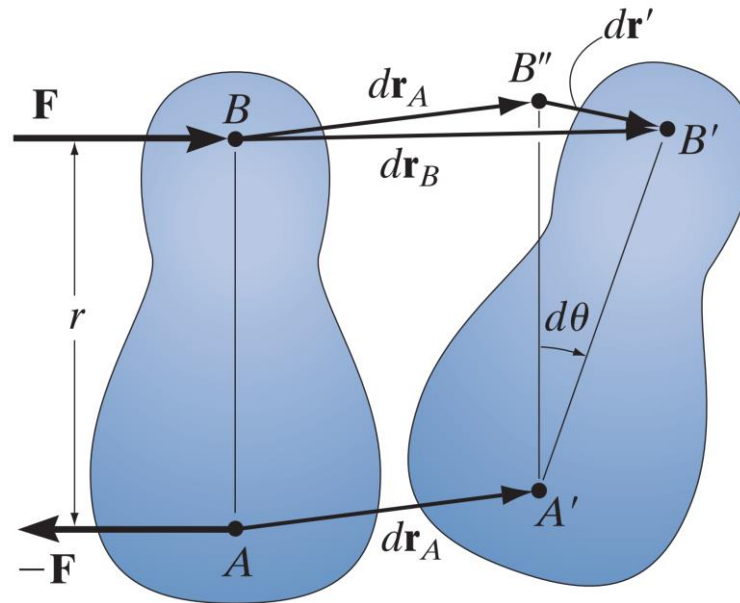
Notice that work is a **scalar**, and like other scalar quantities, it has a magnitude that can either be **positive** or **negative**.

In the *SI* system, the unit of work is a *joule* (J), where $1 \text{ J} = 1 \text{ N} \cdot \text{m}$.

The moment of a force has this same combination of units; however, the concepts of moment and work are in no way related. A moment is a vector quantity, whereas work is a scalar.

Work of a Couple Moment

The rotation of a couple moment also produces work.



$$d\mathbf{r}_B = d\mathbf{r}_A + d\mathbf{r}'$$

$$dU = \mathbf{F} \cdot d\mathbf{r}_B - \mathbf{F} \cdot d\mathbf{r}_A = F \cdot dr'$$

$$dU = Frd\theta = Md\theta$$

Virtual Work

The definitions of the work of a force and a couple have been presented in terms of ***actual movements*** expressed by differential displacements having magnitudes of dr and du .

Consider now an *imaginary* or ***virtual movement*** of a body in static equilibrium. Here the displacement or rotation is *assumed* to occur but *does not actually exist*.

virtual work done by a force:

$$\delta U = F \cos \theta \delta r$$

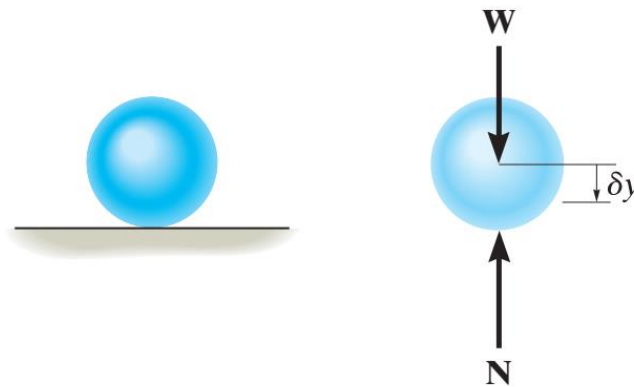
virtual work done by a couple moment:

$$\delta U = M \delta \theta$$

Principle of Virtual Work

The ***principle of virtual work*** states that if a body is in equilibrium, then the algebraic sum of the virtual work done by all the forces and couple moments acting on the body is zero for any virtual displacement of the body. Thus,

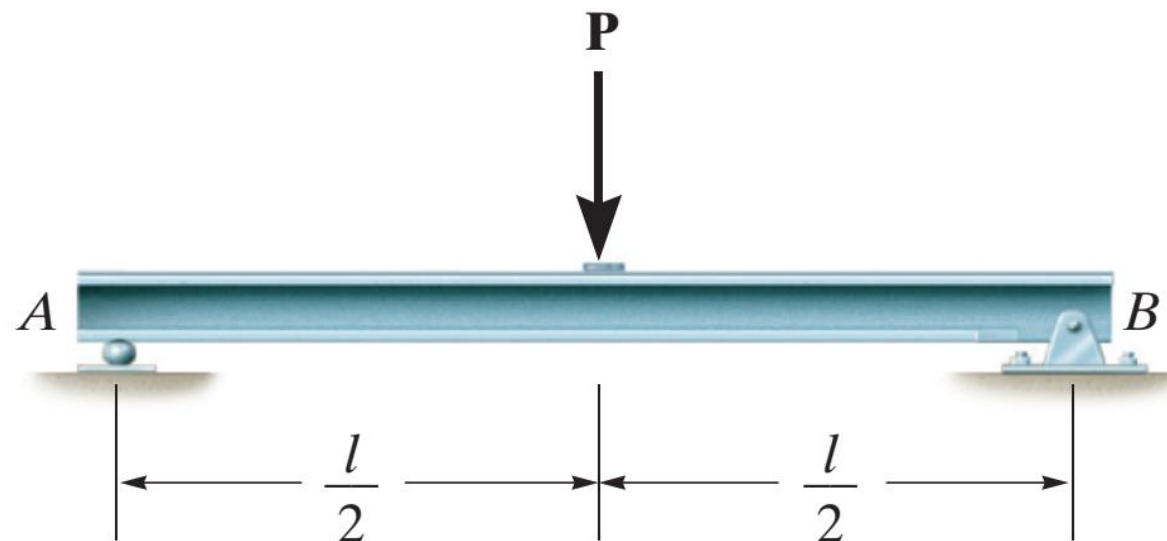
$$\delta U = 0$$



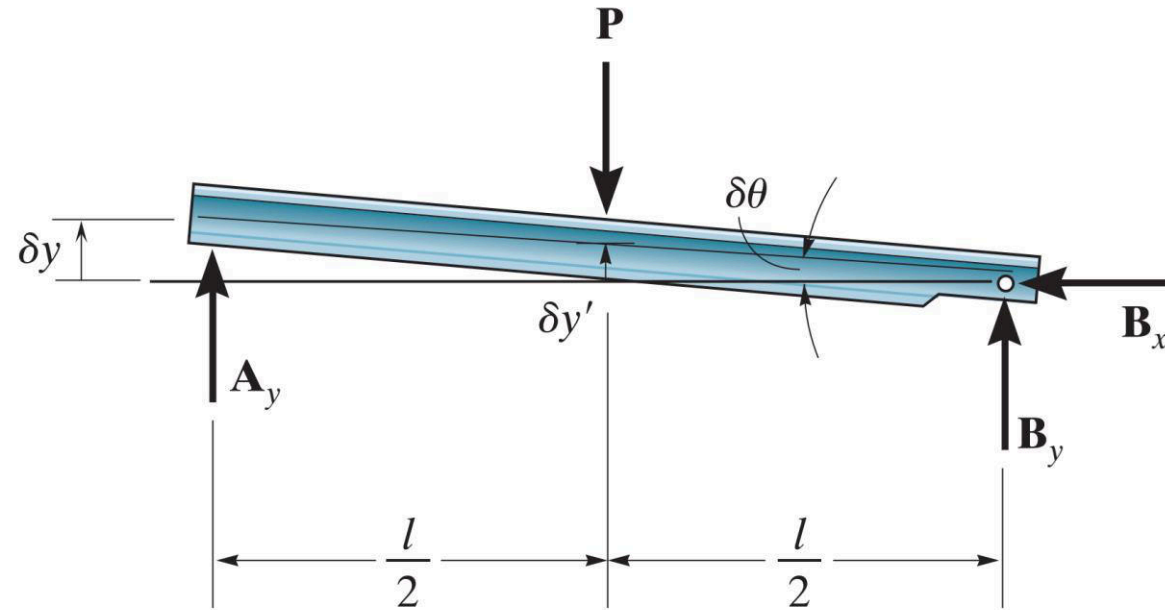
$$\delta U = W \delta y - N \delta y = (W - N) \delta y = 0$$

$$\delta y \neq 0 \quad N = W$$

Principle of Virtual Work



Principle of Virtual Work

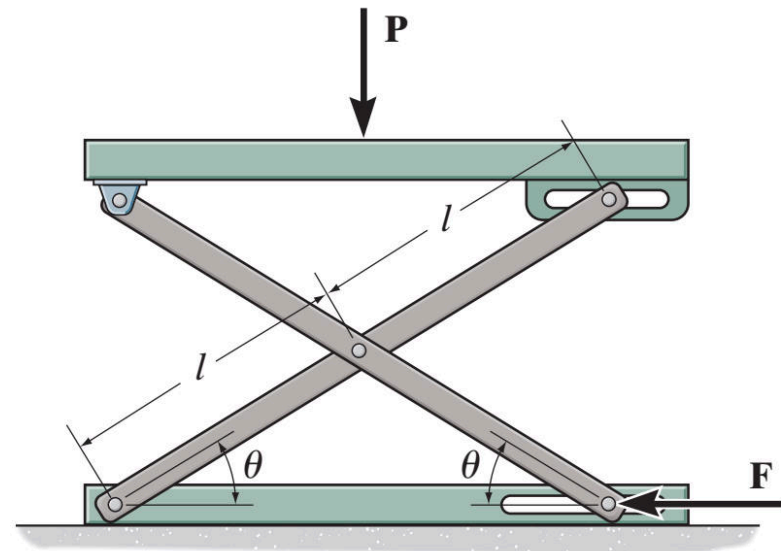
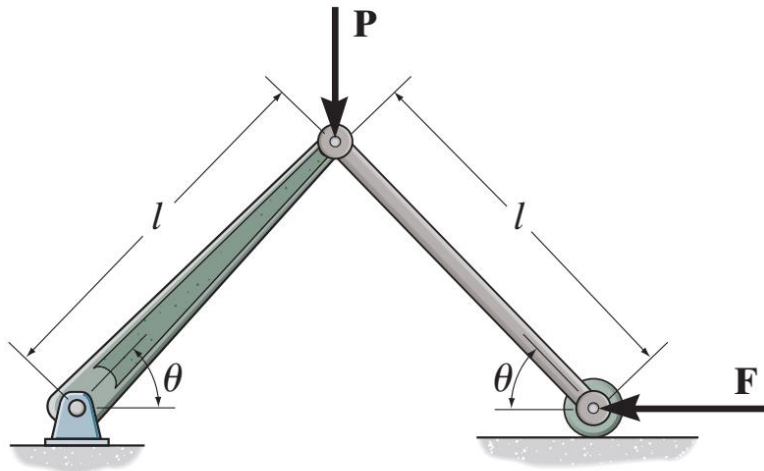


$$\delta U = A_y(l \delta\theta) - P(l/2) \delta\theta = (A_y l - Pl/2) \delta\theta = 0$$

$$\delta\theta \neq 0 \quad A_y = P/2$$

Principle of Virtual Work for A System of Connected Rigid Bodies

The method of virtual work is particularly effective for solving equilibrium problems that involve a system of several *connected* rigid bodies, as shown below:

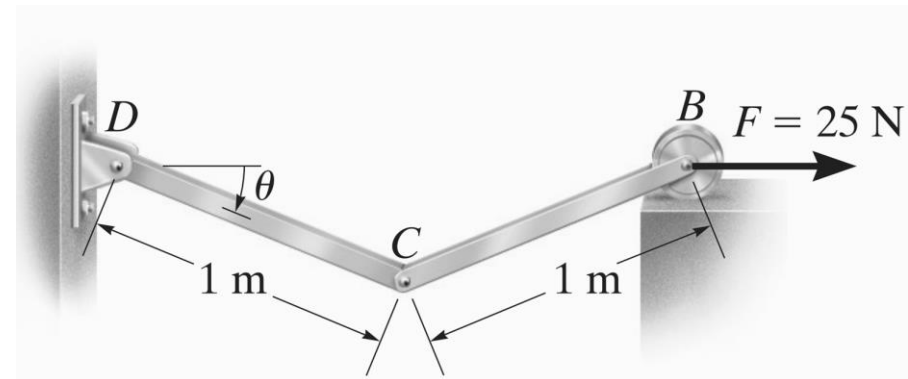


Principle of Virtual Work for A System of Connected Rigid Bodies

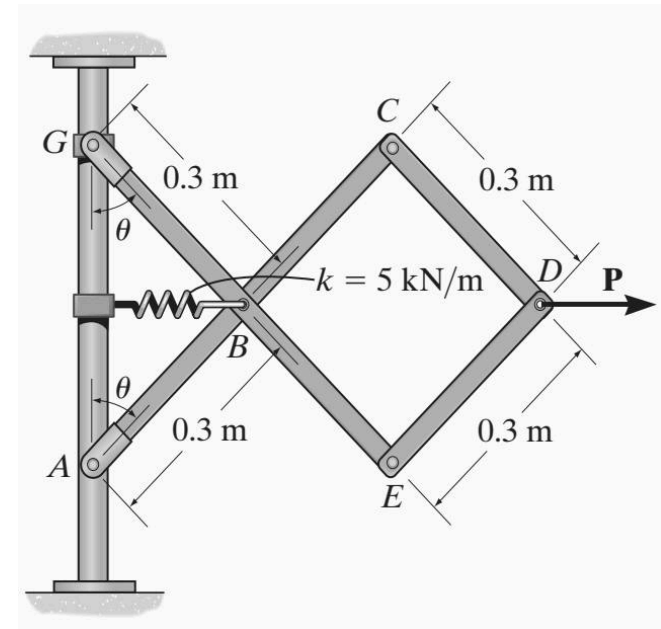


This scissors lift has one degree of freedom. Without the need for dismembering the mechanism, the force in the hydraulic cylinder AB required to provide the lift can be determined *directly* by using the principle of virtual work.

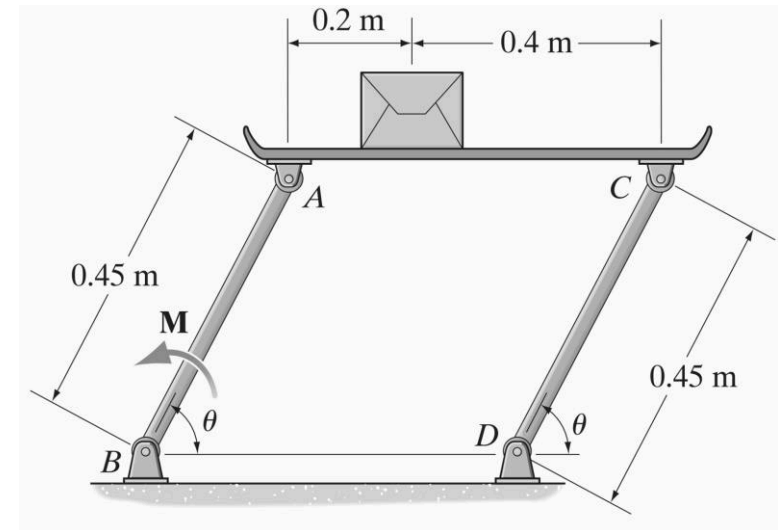
Example: Determine the angle θ for equilibrium of the two-member linkage shown below. Each member has a mass of 10 kg.



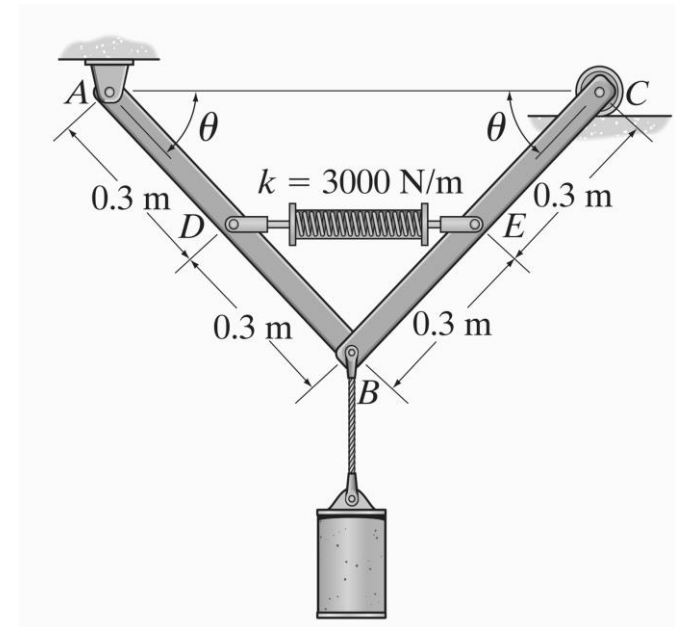
Example: Determine the required force P in figure below needed to maintain equilibrium of the scissors linkage when $\theta = 60^\circ$. The spring is unstretched when $\theta = 30^\circ$. Neglect the mass of the links.



Example: If the box in figure below has a mass of 10 kg, determine the couple moment M needed to maintain equilibrium when $\theta = 60^\circ$. Neglect the mass of the members.



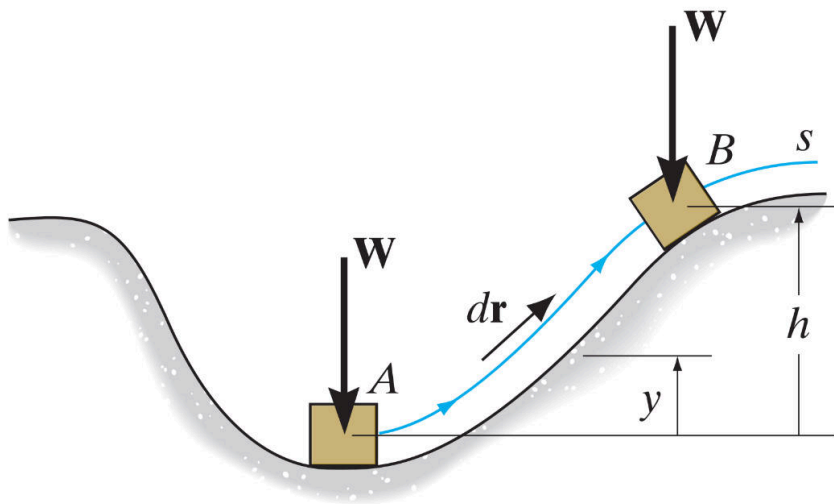
Example: The mechanism below supports the 225-N cylinder. Determine the angle θ for equilibrium if the spring has an unstretched length of 0.6 m when $\theta = 0^\circ$. Neglect the mass of the members.



Conservative Forces

When a force does work that depends only upon the initial and final positions of the force, and it is *independent* of the path it travels, then the force is referred to as a **conservative force**.

Weight



$$U = - \int_0^h W dy = -Wh$$

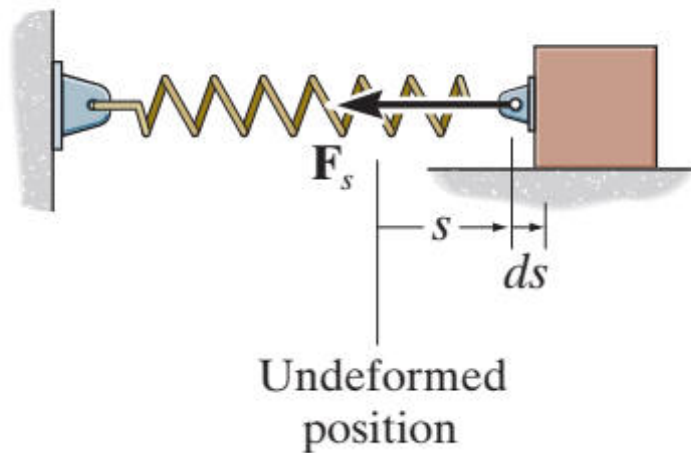
The weight of a body is therefore a conservative force, since the work done by the weight depends only on the *vertical displacement* of the body, and is independent of the path along which the body travels.

Conservative Forces

When a force does work that depends only upon the initial and final positions of the force, and it is *independent* of the path it travels, then the force is referred to as a **conservative force**.

Spring Force

$$U = - \int_{s_1}^{s_2} ks \, ds = - \left(\frac{1}{2} ks_2^2 - \frac{1}{2} ks_1^2 \right)$$



Here the work depends only on the spring's initial and final positions, s_1 and s_2 , measured from the spring's unstretched position. Since this result is independent of the path taken by the block as it moves from s_1 to s_2 , then a spring force is also a *conservative force*.

Conservative Forces

When a force does work that depends only upon the initial and final positions of the force, and it is *independent* of the path it travels, then the force is referred to as a ***conservative force***.

Friction

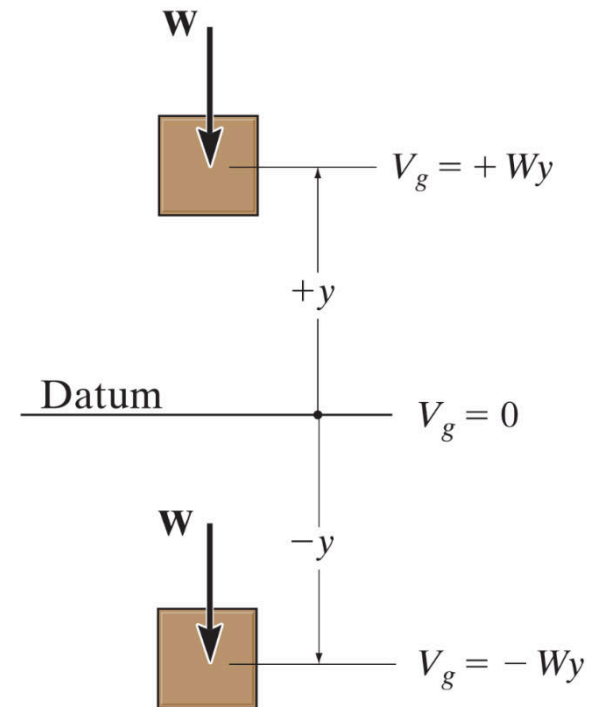
In contrast to a conservative force, consider the force of *friction* exerted on a sliding body by a fixed surface. The work done by the frictional force depends on the path, the longer the path, the greater the work. Consequently, frictional forces are ***nonconservative***, and some of the work done by them is dissipated from the body in the form of heat.

Potential Energy

A conservative force can give the body the **capacity** to do work. This capacity, measured as ***potential energy***, depends on the location or “position” of the body measured relative to a fixed reference position or datum.

Gravitational Potential Energy

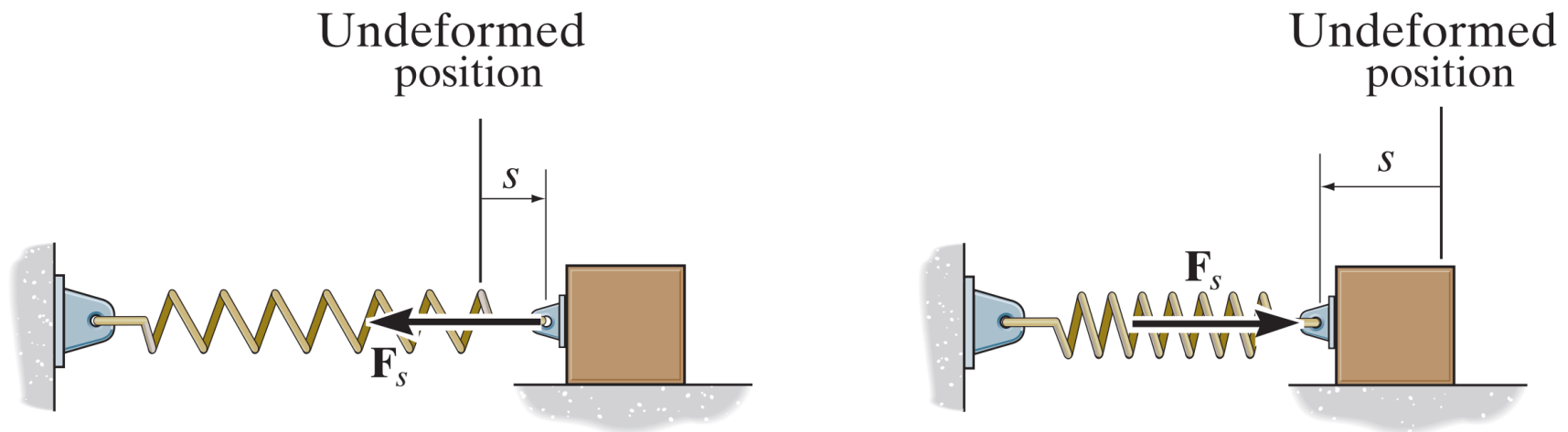
$$V_g = Wy$$



Potential Energy

A conservative force can give the body the **capacity** to do work. This capacity, measured as ***potential energy***, depends on the location or “position” of the body measured relative to a fixed reference position or datum.

Elastic Potential Energy



$$V_e = +\frac{1}{2} k s^2$$

Potential Energy

In the general case, if a body is subjected to *both* gravitational and elastic forces, the *potential energy* or **potential function** V of the body can be expressed as the algebraic sum:

$$V = V_g + V_e$$

If a *system* of frictionless connected rigid bodies (all conservative forces) has a *single degree of freedom* that can be defined by the coordinate q , then the potential function for the system becomes $V = V(q)$. The **work** done by all the weight and spring forces acting on the system as it moves from q_1 to q_2 , is measured by the *difference* in V ;

$$U_{1-2} = V(q_1) - V(q_2)$$

Potential Energy

Work done by conservative forces is equal to the potential energy change in the system:

$$U_{1-2} = V(q_1) - V(q_2)$$

In infinitesimal displacements:

$$dU = V(q) - V(q + dq)$$

$$dU = -dV \quad \text{or} \quad \delta U = -\delta V$$

For equilibrium:

$$\delta U = 0 \quad \text{or} \quad \delta V = 0$$

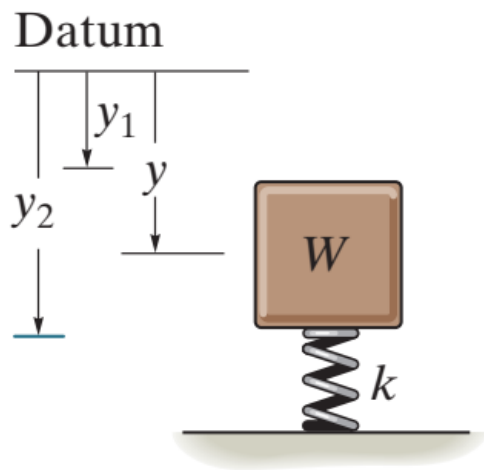
$$\delta V = (dV/dq)\delta q = 0$$

Potential Energy

Thus,

$$\frac{dV}{dq} = 0$$

Hence, *when a frictionless connected system of rigid bodies is in equilibrium, the first derivative of its potential function is equal to zero, or potential energy has an extremum at equilibrium!*

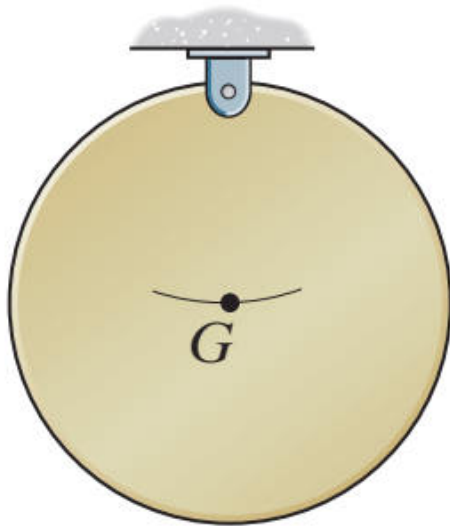


$$\frac{dV}{dy} = -W + ky = 0$$

Types of Equilibrium (Stability)

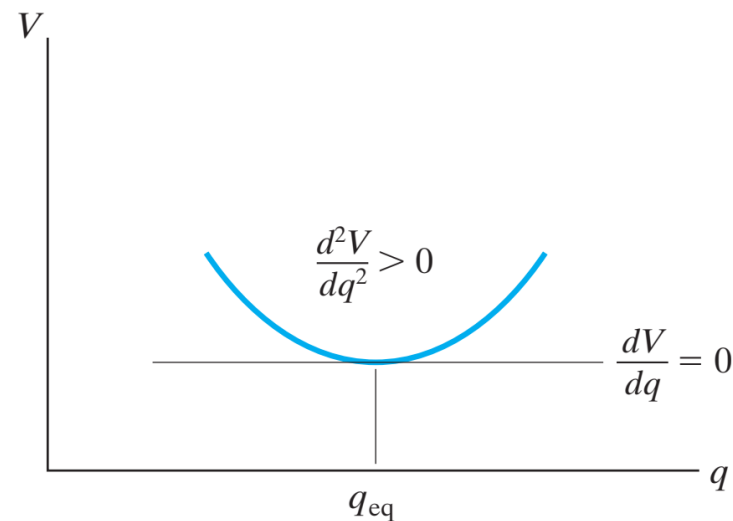
Stable Equilibrium

A system is said to be in ***stable equilibrium*** if the system tends to return to its original position when it is given a small displacement. The potential energy of the system in this case is at its *minimum*.



Stable equilibrium

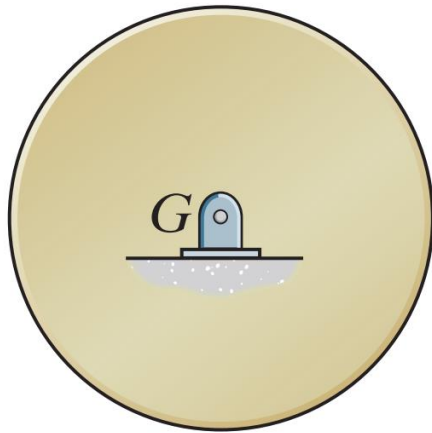
$$\frac{dV}{dq} = 0, \quad \frac{d^2V}{dq^2} > 0$$



Types of Equilibrium (Stability)

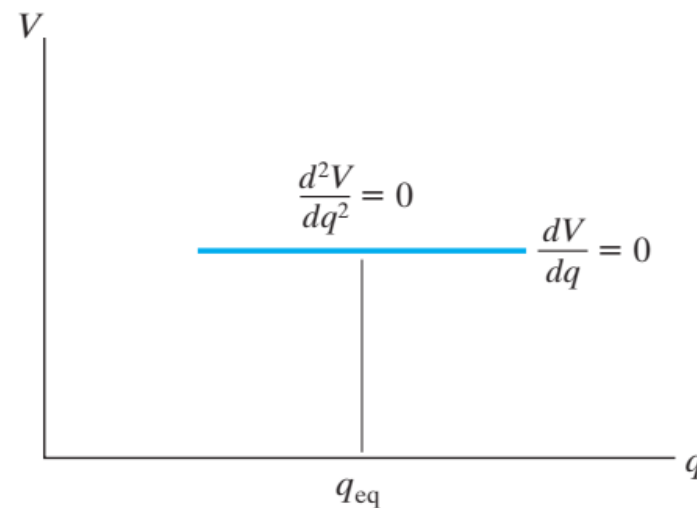
Neutral Equilibrium

A system is said to be in **neutral equilibrium** if the system still remains in equilibrium when the system is given a small displacement away from its original position. In this case, the potential energy of the system is *constant*.



Neutral equilibrium

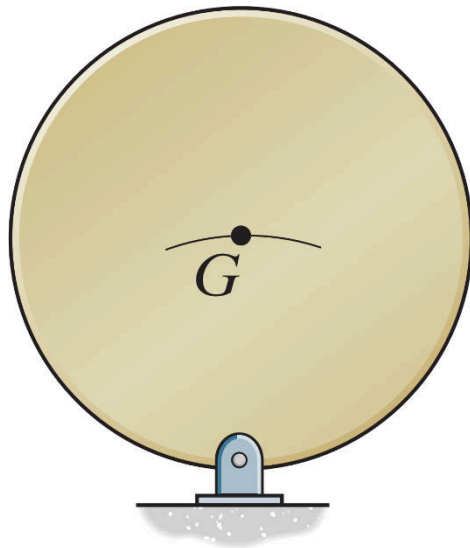
$$\frac{dV}{dq} = \frac{d^2V}{dq^2} = \frac{d^3V}{dq^3} = \dots = 0$$



Types of Equilibrium (Stability)

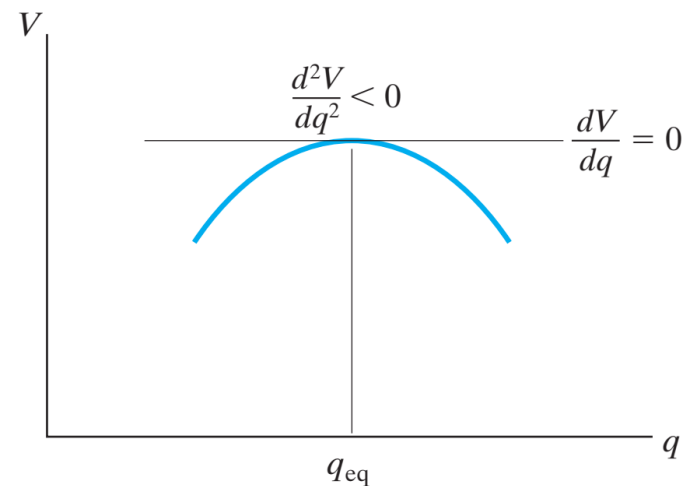
Unstable Equilibrium

A system is said to be in ***unstable equilibrium*** if it tends to be *displaced farther away* from its original equilibrium position when it is given a small displacement. The potential energy of this system is maximum.

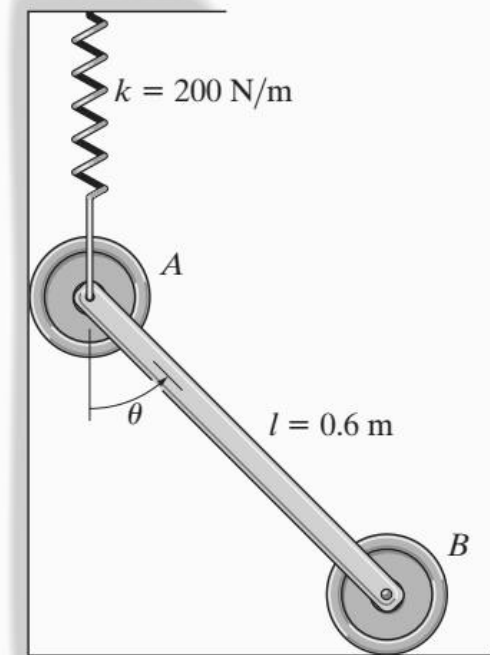


Unstable equilibrium

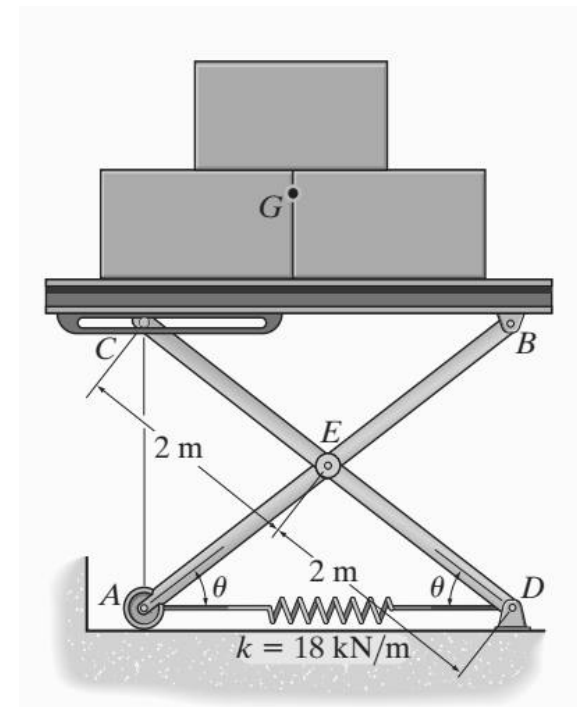
$$\frac{dV}{dq} = 0, \quad \frac{d^2V}{dq^2} < 0$$



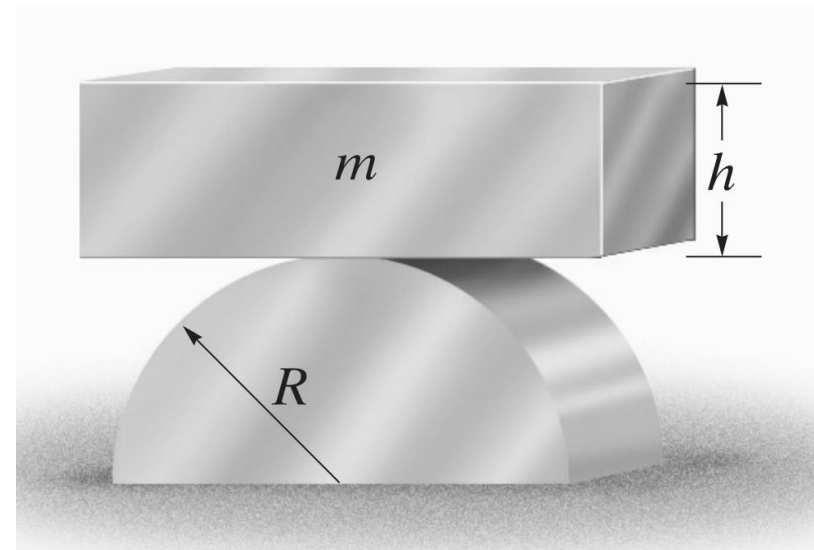
Example: The uniform link shown below has a mass of 10 kg. If the spring is unstretched when $\theta = 0^\circ$, determine the angle θ for equilibrium and investigate the stability at the equilibrium position.



Example: If the spring AD shown below has a stiffness of 18 kN/m and is unstretched when $\theta = 60^\circ$, determine the angle θ for equilibrium. The load has a mass of 1.5 Mg . Investigate the stability at the equilibrium position.

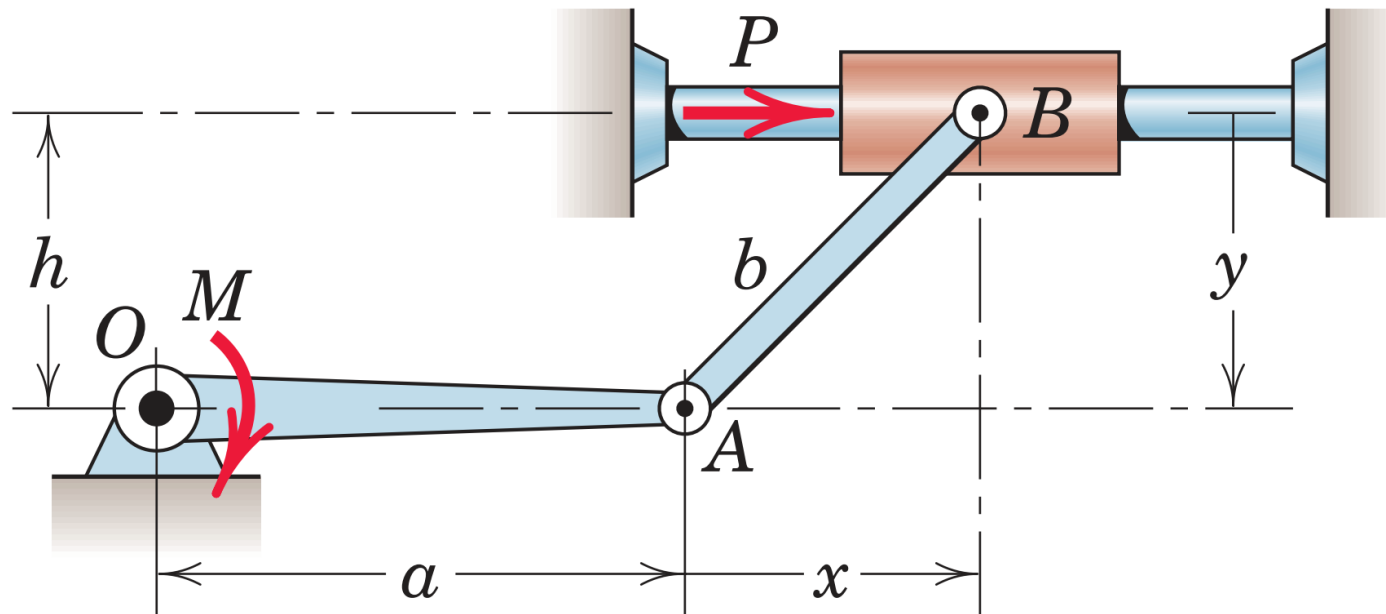


Example: The uniform block having a mass m rests on the top surface of the half cylinder. Show that this is a condition of unstable equilibrium if $h > 2R$.



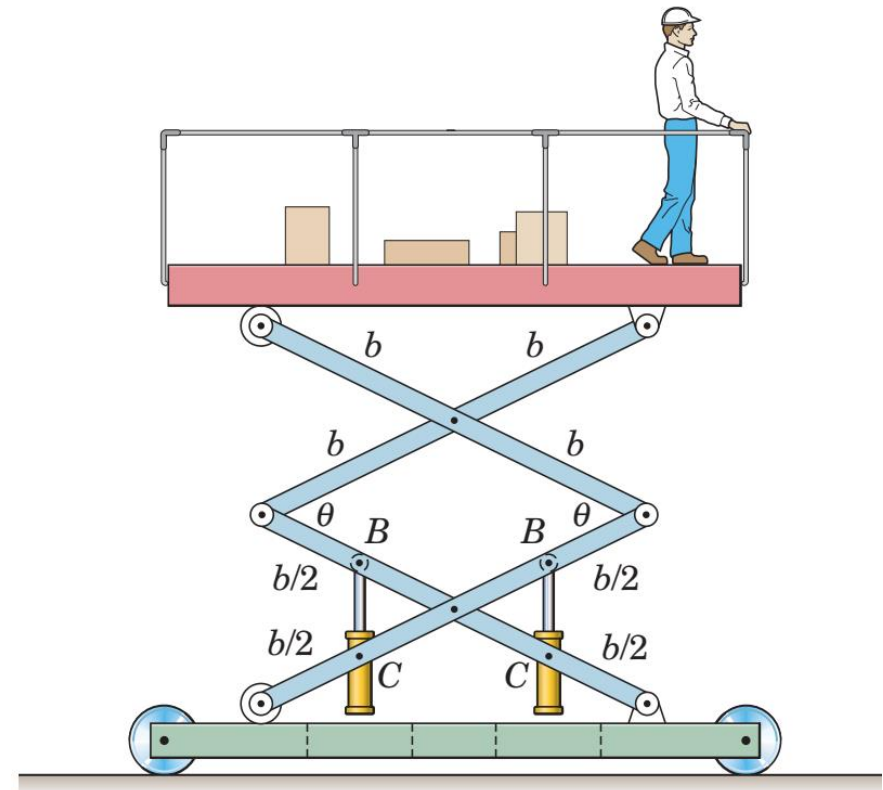
Practice at Home

1. For link OA in the horizontal position shown, determine the force P on the sliding collar which will prevent OA from rotating under the action of the couple M . Neglect the mass of the moving parts.



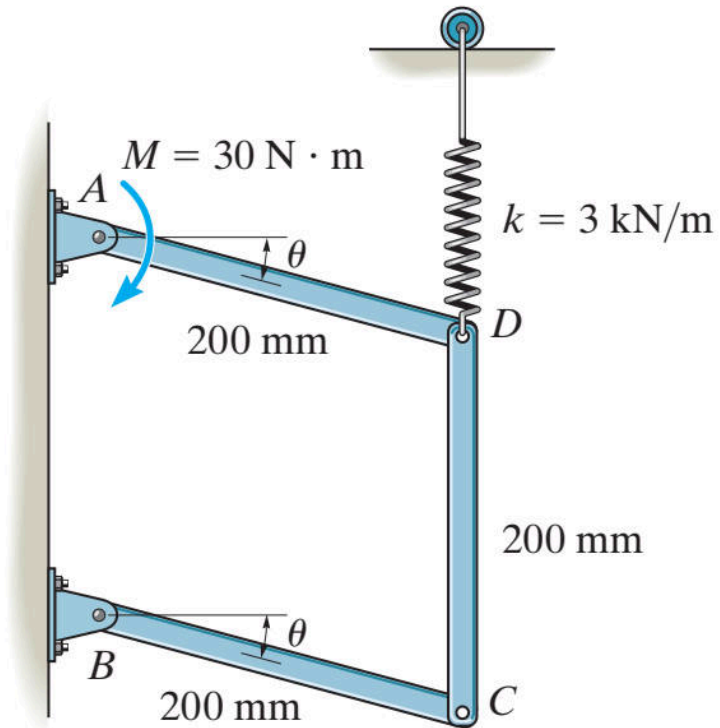
Practice at Home

2. The portable work platform is elevated by means of the two hydraulic cylinders articulated at points C . Each cylinder is under a hydraulic pressure p and has a piston area A . Determine the pressure p required to support the platform and show that it is independent of θ . The platform, worker, and supplies have a combined mass m , and the masses of the links may be neglected.



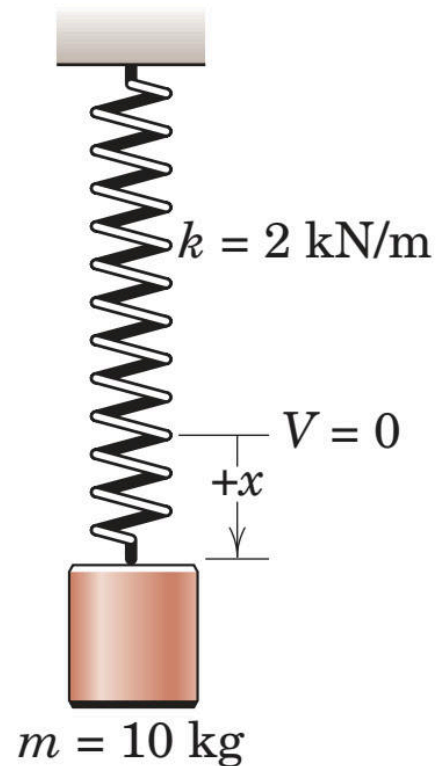
Practice at Home

3. If each of the three links of the mechanism has a mass of 4 kg, determine the angle θ for equilibrium. The spring, which always remains vertical, is unstretched when $\theta = 0^\circ$.



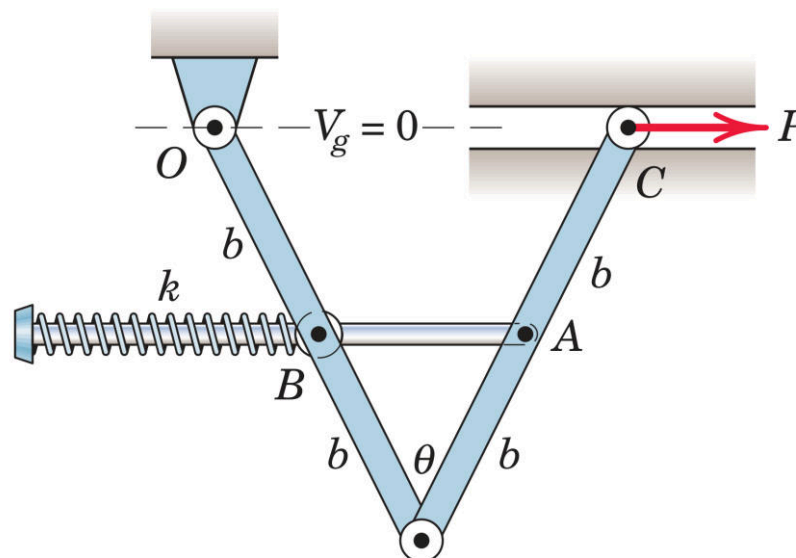
Practice at Home

4. The 10-kg cylinder is suspended by the spring, which has a stiffness of 2 kN/m. Plot the potential energy V of the system and show that it is minimum at the equilibrium position.



Practice at Home

5. The two uniform links, each of mass m , are in the vertical plane and are connected and constrained as shown. As the angle θ between the links increases with the application of the horizontal force P , the light rod, which is connected at A and passes through a pivoted collar at B , compresses the spring of stiffness k . If the spring is uncompressed in the position where $\theta = 0$, determine the force P which will produce equilibrium at the angle θ .



Practice at Home

6. The ends of the uniform bar of mass m slide freely in the horizontal and vertical guides. Examine the stability conditions for the positions of equilibrium. The spring of stiffness k is undeformed when $x = 0$.

